Solvent-Based Adhesives

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Solvent Based Contact Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength of Adhesive Bonds); ISO 8510 (Peel Adhesion); ISO 10365 (Adhesive Failure Classification),ISO 10364; ISO 3251	ASTM D1002 (Strength Properties of Adhesively Bonded Metal Lap Joints); ASTM D3165 (Strength Properties of Adhesives in Shear by Tension Loading); ASTM D903 (Peel or Stripping Strength); ASTM D1876 (T-Peel Resistance of Adhesives); ASTM D3330 (Peel Adhesion of Pressure Sensitive Materials where applicable),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1465 (Adhesives — Determination of Tensile Lap-Shear Strength); DIN EN 13999 series (Test Methods for Adhesives — VOC Emissions); DIN EN ISO 9142 (Guidelines for Adhesive Bonding Durability); DIN EN ISO 11339 (T-Peel Test for Adhesives); DIN EN ISO 10365 (Designation of Main Failure Patterns)	REACH: Regulation (EC) No. 1907/2006 (solvents, VOC, SVHC substances); REACH Annex XVII (restricted solvents); FDA: 21 CFR 175.105 where used in food-contact packaging adhesive applications
Neoprene Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 8510 (Peel Strength); ISO 10365 (Failure Classification),ISO 10364; ISO 3251	ASTM D429 (Rubber Property—Adhesion to Rigid Substrates); ASTM D1002 (Lap Shear Strength); ASTM D903 (Peel Strength); ASTM D3165 (Adhesive Shear Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 9142 (Durability of Adhesive Bonds); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (T-Peel Strength); DIN EN ISO 10365 (Adhesive Failure Classification)	REACH: Regulation (EC) No. 1907/2006 (chloroprene rubber, solvents, VOC); REACH Annex XVII restrictions on hazardous substances; FDA: 21 CFR 175.105 where food-contact use applies
Nitrile Rubber Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 11339 (T-Peel Test); ISO 10365,ISO 10364; ISO 3251	ASTM D429 (Rubber Adhesion Testing); ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D903 (Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 9142 (Durability); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 527 series (Mechanical Properties where polymer film testing applies)	REACH: Regulation (EC) No. 1907/2006 (nitrile polymers, curing agents, solvents); FDA: 21 CFR 175.105 where applicable
Polyurethane Solvent Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 11339 (T-Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D1876 (T-Peel Strength); ASTM D903 (Peel Strength); ASTM D3762 (Adhesive Bond Strength of Aluminium where applicable),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1465 (Tensile Lap-Shear Strength); DIN EN ISO 11339 (T-Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Mode Classification)	REACH: Regulation (EC) No. 1907/2006 (isocyanates, solvents, VOC); REACH Annex XVII (diisocyanate restrictions); FDA: 21 CFR 175.105 where used for food-contact packaging
Acrylic Solvent Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 8510 (Peel Adhesion); ISO 10365,ISO 10364; ISO 3251	ASTM D1002 (Lap Shear Strength); ASTM D903 (Peel Strength); ASTM D3330 (Peel Adhesion); ASTM D3654 (Shear Adhesion Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 11339 (T-Peel Strength); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 9142 (Environmental Durability); DIN EN ISO 10365 (Adhesive Failure Pattern)	REACH: Regulation (EC) No. 1907/2006 (acrylate monomers, solvents, VOC); FDA: 21 CFR 175.105 where applicable

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Rubber Based Adhesive	IS 4643; IS 8542	ISO 4587 (Adhesive Bond Strength); ISO 11339 (Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM D429 (Rubber Adhesion); ASTM D903 (Peel Strength); ASTM D1002 (Shear Strength); ASTM D3165 (Adhesive Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 9142 (Durability of Adhesive Bonds); DIN EN 1465 (Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 527 (Mechanical Properties where applicable)	REACH: Regulation (EC) No. 1907/2006 (rubber additives, accelerators, solvents, VOC); FDA: 21 CFR 175.105 where food-contact applications exist
Shoe Adhesive	IS 4643; IS 8542	ISO 17707 (Footwear Sole Adhesion); ISO 4587 (Tensile Shear Strength); ISO 8510 (Peel Adhesion),ISO 10364; ISO 3251	ASTM D2209 (Adhesion of Leather and Shoe Materials where applicable); ASTM D903 (Peel Strength); ASTM D1002 (Lap Shear Strength); ASTM D1876 (T-Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1392 (Adhesives for Leather and Footwear Materials — Testing Methods); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (solvents, VOC, restricted substances); REACH Annex XVII restrictions on hazardous chemicals
Leather Adhesive	IS 4643; IS 8542	ISO 11640 (Leather Testing where applicable); ISO 4587; ISO 8510,ISO 10364; ISO 3251	ASTM D2209 (Strength Properties of Leather); ASTM D903 (Peel Strength); ASTM D1002 (Lap Shear Strength); ASTM D3165 (Adhesive Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1392 (Adhesives for Leather and Footwear Materials); DIN EN ISO 11640 (Leather Colour Fastness where relevant); DIN EN 1465 (Shear Strength); DIN EN ISO 11339 (Peel Strength)	REACH: Regulation (EC) No. 1907/2006 (solvents, dyes, restricted substances); REACH Annex XVII (restricted azo compounds where applicable)
Furniture Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 11339 (Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM D905 (Strength Properties of Adhesive Bonds in Shear by Compression Loading); ASTM D1002 (Lap Shear Strength); ASTM D1876 (T-Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897	DIN EN 204 (Classification of Thermoplastic Wood Adhesives); DIN EN 205 (Wood Adhesives — Determination of Tensile Shear Strength); DIN EN 14257 (WATT 91 Heat Resistance Test); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (formaldehyde, VOC, solvents, additives); FDA: —
Lamination Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 11339 (T-Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM D1876 (T-Peel Strength); ASTM D903 (Peel Strength); ASTM D1002 (Lap Shear Strength); ASTM F88 (Seal Strength of Flexible Barrier Materials where packaging lamination applies),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (T-Peel Strength); DIN EN ISO 527 (Mechanical Properties of Polymer Films); DIN EN 1939 (Bond Strength of Adhesive Tapes and Laminates where applicable)	REACH: Regulation (EC) No. 1907/2006 (solvents, isocyanates, VOC, additives); FDA: 21 CFR 175.105 where used in food packaging laminates
Packaging Adhesive	IS 4643; IS 8542	ISO 4587 (Adhesive Bond Strength); ISO 11339 (Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM D903 (Peel Strength); ASTM D1002 (Lap Shear Strength); ASTM F88 (Seal Strength of Flexible Barrier Materials); ASTM D1876 (T-Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1895 (Peel Strength of Adhesive Bonds); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 527 (Film Mechanical Properties where laminated packaging applies)	FDA: 21 CFR 175.105 (Adhesives); FDA: 21 CFR 176.170 / 176.180 where used in food-contact paper/paperboard structures; REACH: Regulation (EC) No. 1907/2006

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Automotive Adhesive	IS 4643; IS 8542	ISO 16750 (Road Vehicle Environmental Testing where applicable); ISO 4587; ISO 11343 (Impact Wedge Peel Test),ISO 10364; ISO 3251	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D3762 (Adhesive Bond Strength of Aluminium); ASTM D1876 (Peel Strength); ASTM D429 (Rubber Adhesion where applicable),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 9142 (Durability of Adhesive Bonds); DIN EN 1465 (Lap Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 10365 (Failure Classification); DIN EN ISO 16701 (Corrosion Protection Evaluation where applicable)	REACH: Regulation (EC) No. 1907/2006 (isocyanates, solvents, restricted substances); REACH Annex XVII (diisocyanates where PU systems apply)
Construction Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 11339 (Peel Strength); ISO 10365,ISO 10364; ISO 3251	ASTM C557 (Adhesives for Installation of Gypsum Wallboard); ASTM D1002 (Lap Shear Strength); ASTM D903 (Peel Strength); ASTM D4541 (Pull-Off Adhesion for Coated Surfaces where applicable),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 15651 series (Sealants for Building Construction); DIN EN 1465 (Adhesive Strength); DIN EN ISO 11600 (Classification of Sealants); DIN EN ISO 9142 (Durability)	REACH: Regulation (EC) No. 1907/2006 (VOC, solvents, formaldehyde, additives); REACH Annex XVII restrictions
Metal Bonding Adhesive	IS 4643; IS 8542	ISO 4587 (Tensile Shear Strength); ISO 10365 (Failure Classification); ISO 12944 where corrosion protection applies,ISO 10364; ISO 3251	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Adhesive Shear Strength); ASTM D3762 (Aluminium Bond Strength); ASTM D903 (Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1465 (Tensile Lap-Shear Strength); DIN EN ISO 11339 (Peel Strength); DIN EN ISO 9142 (Durability); DIN EN ISO 10365 (Failure Modes)	AMPP: SP 1/SP 2/SP 3 where bonded metal surface preparation applies; REACH: Regulation (EC) No. 1907/2006 (epoxy, PU, solvents, additives)
Wood Bonding Adhesive	IS 4835:1979 (PVAc) / IS 848:2006 (phenolic & aminoplastic); IS 4643; IS 8542	ISO 12465 (Wood Adhesive Bonding Applications where applicable); ISO 4587; ISO 11339,ISO 10364; ISO 3251	ASTM D905 (Strength Properties of Adhesive Bonds in Shear by Compression Loading); ASTM D1002 (Lap Shear Strength); ASTM D1876 (T-Peel Strength); ASTM D5751 (Adhesives Used in Wood Products where applicable),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897	DIN EN 204 (Thermoplastic Wood Adhesives Classification); DIN EN 205 (Wood Adhesive Shear Strength); DIN EN 14257 (WATT 91 Heat Resistance); DIN EN 14256 (Open Assembly Time)	REACH: Regulation (EC) No. 1907/2006 (formaldehyde, VOC, additives); FDA: 21 CFR 175.105 only for food-contact wood applications
PVC Solvent Cement	IS 14182:1994,IS 4643; IS 8542	ISO 15853 (Solvent Cements for Thermoplastics where applicable); ISO 4587; ISO 8510,ISO 10364; ISO 3251	ASTM D2564 (Solvent Cements for Polyvinyl Chloride Pipe and Fittings); ASTM D3164 (Strength Properties of Adhesively Bonded Plastic Joints); ASTM D2855 (Making Solvent-Cemented Joints with PVC Pipe and Fittings),ASTM D1002; ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 14680 (Adhesives for Non-Pressure Thermoplastic Piping Systems); DIN EN 14814 (Adhesives for Thermoplastic Piping Systems for Fluids under Pressure); DIN EN ISO 19069 series (Thermoplastics Testing where applicable)	REACH: Regulation (EC) No. 1907/2006 (solvents, PVC additives, VOC); FDA: 21 CFR 175.105 where used in food-contact piping applications
Pipe Joint Adhesive	IS 14182:1994 (for PVC-U solvent cement),IS 4643;	ISO 15853 (Pipe Jointing Materials where applicable); ISO 4587; ISO 10365,ISO 10364; ISO 3251	ASTM D2855 (PVC Solvent Cement Joint Practice); ASTM D2564 (PVC Solvent Cement); ASTM D3164 (Plastic	DIN EN 14814 (Adhesives for thermoplastic piping systems for fluids under pressure —	REACH: Regulation (EC) No. 1907/2006 (solvents, VOC, PVC/CPVC additives); FDA: applicable for potable

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
	IS 8542		Adhesive Joint Strength),ASTM D1002; ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	Specifications); DIN EN 14680 (Adhesives for non-pressure thermoplastic piping systems — Specifications); DIN EN ISO 19069 series (Thermoplastics — Mechanical properties where pipe substrate testing is required); DIN EN 923 (Adhesives — Terms and definitions); DIN EN 1465 (Adhesives — Determination of tensile lap-shear strength); DIN EN ISO 9142 (Adhesives — Guide to selection of standard laboratory ageing conditions); DIN EN ISO 11339 (Adhesives — T-peel test for flexible-to-flexible bonded assemblies)	water/food-contact piping adhesive systems where compliant formulations are used
Spray Adhesive	IS 4643; IS 8542	ISO 4587 (Bond Strength); ISO 8510 (Peel Adhesion); ISO 10365,ISO 10364; ISO 3251	ASTM D903 (Peel Strength); ASTM D1002 (Lap Shear Strength); ASTM D3330 (Peel Adhesion); ASTM D3654 (Shear Adhesion),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN 1465 (Tensile lap-shear strength); DIN EN ISO 11339 (T-peel strength); DIN EN ISO 9142 (Durability of adhesive bonds); DIN EN ISO 10365 (Designation of main failure patterns); DIN EN 13999 series (Test methods for VOC emissions from adhesives); DIN EN ISO 16000 series (Indoor air quality — VOC measurement where used in construction/interior applications)	REACH: Regulation (EC) No. 1907/2006 (propellants, solvents, VOC, polymers); REACH Annex XVII restrictions on hazardous solvents; FDA: 21 CFR 175.105 where food-contact application exists
Pressure Sensitive Solvent Adhesive	IS 4643; IS 8542	ISO 29862 (Peel Adhesion); ISO 29863 (Shear Adhesion); ISO 29864 (Tensile Strength),ISO 10364; ISO 3251	ASTM D3330 (Peel Adhesion of Pressure Sensitive Tape); ASTM D3654 (Shear Adhesion); ASTM D2979 (Tack of Pressure Sensitive Adhesives); ASTM D6252 (Peel Adhesion of Pressure Sensitive Labels),ASTM D1002; ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 29862 (Self adhesive tapes — Determination of peel adhesion properties); DIN EN ISO 29863 (Self adhesive tapes — Determination of static shear adhesion); DIN EN ISO 29864 (Self adhesive tapes — Measurement of breaking strength and elongation); DIN EN ISO 29861 (Self adhesive tapes — Measurement of quick stick/tack); DIN EN 1939 (Determination of peel adhesion properties of adhesive tapes); DIN EN ISO 9142 (Durability of adhesive bonds); DIN EN 13999 series (VOC emissions from adhesives)	REACH: Regulation (EC) No. 1907/2006 (solvents, tackifiers, acrylic/rubber components, VOC); FDA: 21 CFR 175.105 where food-contact labels/tapes are used
Heat Resistant Solvent Adhesive	IS 4643; IS 8542	ISO 4587 (Bond Strength); ISO 10365; ISO 527 (Mechanical Properties where applicable),ISO 10364; ISO 3251	ASTM D1002 (Lap Shear Strength); ASTM D3165 (Shear Strength); ASTM D2294 (Adhesive Bond Strength after Exposure to Heat where applicable); ASTM D903 (Peel Strength),ASTM D1084; ASTM D1151; ASTM D1489; ASTM D1557; ASTM D1993; ASTM D2556; ASTM D3236; ASTM D4498; ASTM D4499; ASTM D897; ASTM D905	DIN EN ISO 9142 (Adhesive bond durability under environmental conditions); DIN EN 1465 (Tensile lap-shear strength); DIN EN ISO 11339 (T-peel strength); DIN EN ISO 10365 (Failure pattern classification); DIN EN ISO 2812 series (Resistance to liquids including chemicals/solvents); DIN EN ISO 3248 (Effect of heat on	REACH: Regulation (EC) No. 1907/2006 (high-temperature resins, solvents, additives); FDA: only where used in food-contact applications

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
				coatings/adhesive systems where applicable); DIN EN ISO 527 series (Mechanical properties of polymer-based adhesive systems where applicable)	

Source field: Dyes and Pigments